

Advanced Machine Learning

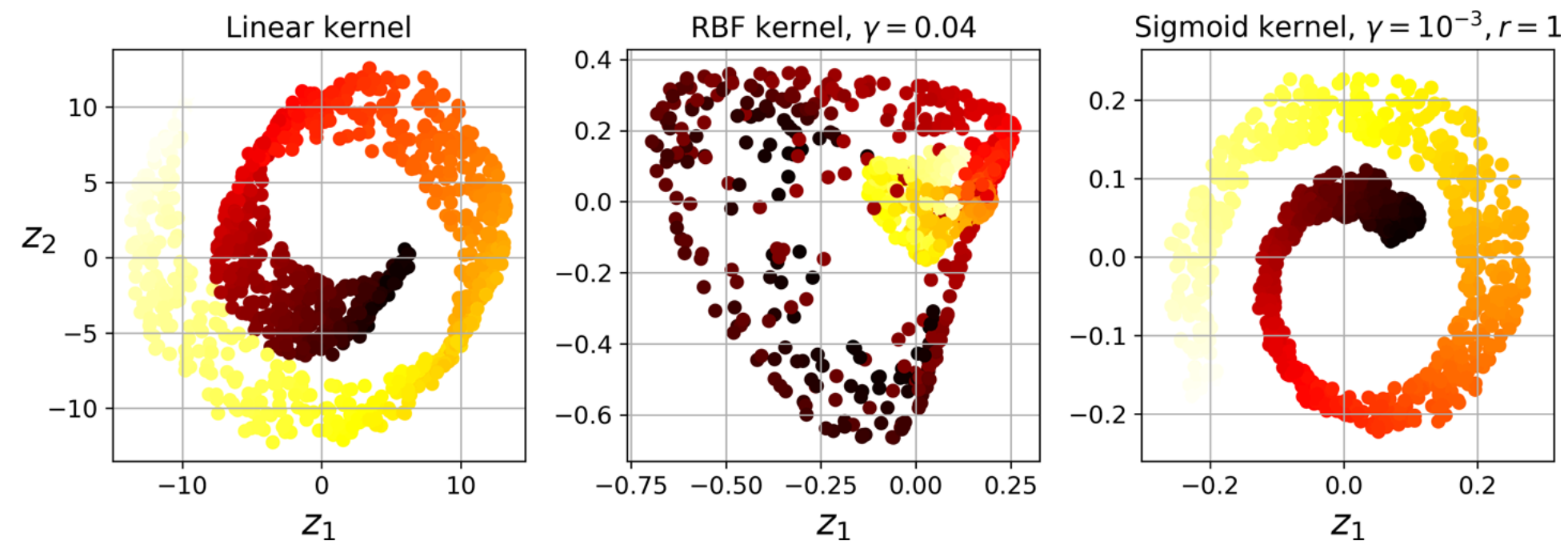
DATA 642

Lecture 6 — Kernel PCA

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Kernel PCA

- The kernel trick can be applied to PCA, making it possible to perform complex nonlinear projections for dimensionality reduction. This is called **Kernel PCA (kPCA)**.
- Kernel PCA computes the eigenvectors of the Gram matrix constructed from the kernel function applied to all pairs of data points, which represent the principal components in the feature space induced by the kernel function. These eigenvectors are known as the kernel principal components.



Selecting a Kernel and Tuning Hyperparameters via supervised task

- As kPCA is an unsupervised learning algorithm, there is no obvious performance measure to help you select the best kernel and hyperparameter values.
- Dimensionality reduction is often a preparation step for a supervised learning task (e.g., classification), so you can use grid search to select the kernel and hyperparameters that lead to the best performance on that task.

Selecting a Kernel and Tuning Hyperparameters via reconstruction error

- Another approach, entirely unsupervised, is to select the kernel and hyperparameters that yield the lowest reconstruction error.
- Reconstruction error in kPCA is not as straightforward as in linear PCA due to the use of the kernel trick.
- The transformation in kPCA using a kernel is mathematically equivalent to mapping the training set to an infinite-dimensional feature space and then projecting it down to the desired dimension using linear PCA.
- Reconstruction error is measured using a reconstruction pre-image, which is a point in the original space that maps close to the reconstructed point.
- The reconstruction pre-image error is minimized to select the kernel and hyperparameters.

